



Habib University

Digital Logic and Design (EE-172 / CS-130 / CE-222)

Fall 2025

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Assignment No: 4

Release Date: November 12th, 2025

Due by: November 19th, 2025 11:59 PM

Total points: 100

Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; parity bits; representing complements of a number; and performing BCD arithmetic.

Instructions:

This assignment should be done individually.

All questions should be answered in **black ink only**.

Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

Your assignments will be checked by instructor/TA.

You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.

Zero will be given if the assignment is found to be plagiarized.

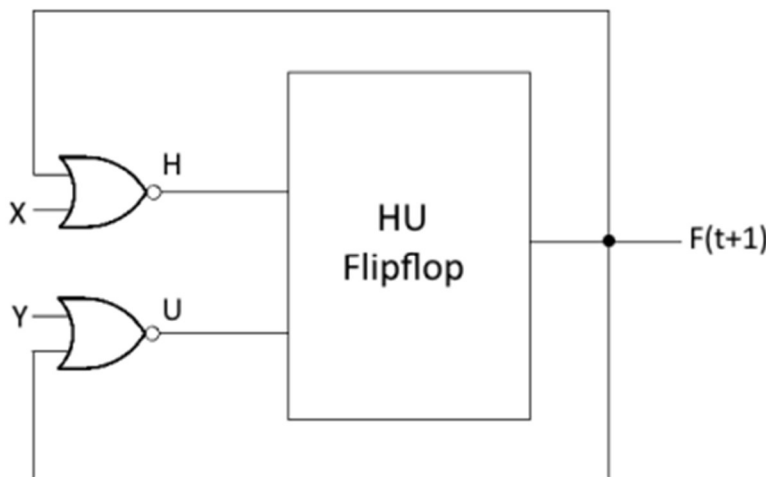
Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:

1-day late submission – 20% deduction of the maximum allowable marks

2-days late submission – 40% deduction of the maximum allowable marks

No submission will be accepted after two days of the original deadline

Q#	Questions	Pts																																				
1	Consider the following sequential circuit:  The circuit has two inputs X and Y, and one state variable F. H and U represent the outputs at indicated points in circuit diagram. The circuit requires a hypothetical flip that computes the next state of F using H and U in the following manner: a. Clear output to 0 when $HU = 11$ b. Complement current output when $HU = 10$ c. Set output to 1 when $HU = 01$, and d. No change when $HU = 00$ You can right away identify there is no such flip flop in the world, so you decide to define one of your own! a. Draw the characteristic table for your new flip flop HU. <table data-bbox="449 1348 924 1740"><thead><tr><th>F(t)</th><th>H</th><th>U</th><th>F(t+1)</th></tr></thead><tbody><tr><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td></tr></tbody></table> b. Derive the characteristic equation. $F(t+1) = y = H'U + FH'U' + F'HU' \text{ or } F'H'U + F'HU' + FH'U' + FH'U$	F(t)	H	U	F(t+1)	0	0	0	0	0	0	1	1	0	1	0	1	0	1	1	0	1	0	0	1	1	0	1	1	1	1	0	0	1	1	1	0	2x3+4
F(t)	H	U	F(t+1)																																			
0	0	0	0																																			
0	0	1	1																																			
0	1	0	1																																			
0	1	1	0																																			
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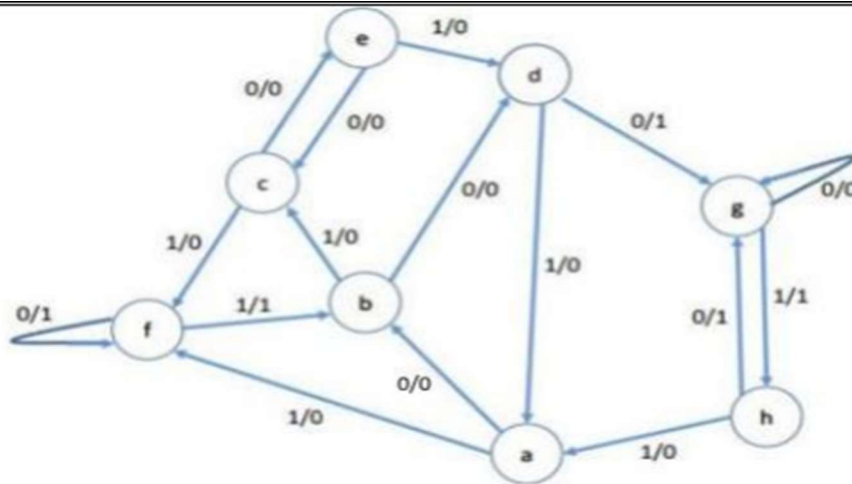
c. Tabulate the excitation table.

F(t)	F(t+1)	H	U
0	0	X	H
0	1	X	H'
1	0	1	X
1	1	0	X

d. Obtain the state transition table for the given circuit.

Current State	Inputs		Next State	Output	
F(t)	X	Y	F(t+1)	H	U
0	0	0	0	1	1
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	0	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

2



3+8+10
+3

a. Draw the respective state table.

Previous State	Next State		Output	
	X = 0	X = 1	X = 0	X = 1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

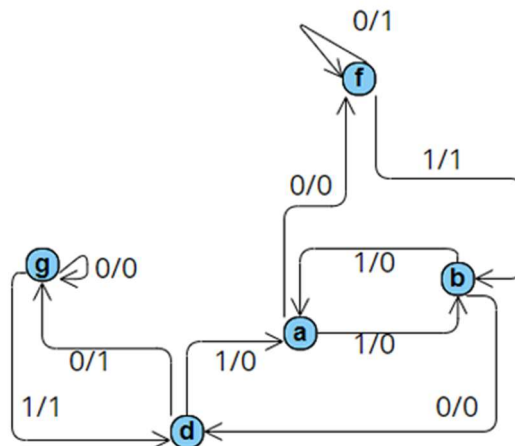
- b. Reduce the state diagram by finding equivalent states in state table rows.

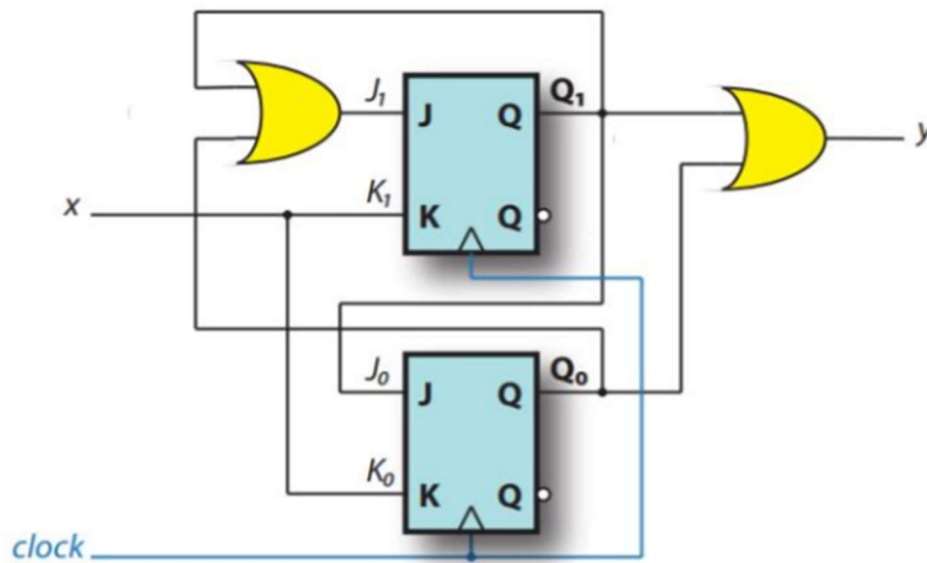
Previous State	Next State		Output	
	X = 0	X = 1	X = 0	X = 1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
f	f	b	1	1
g	g	h	0	1

- c. Reduce the state diagram using the Implication Chart.

b	x	-	-	-	-	-	-
c	f-f e-b	f-d c-e	-	-	-	-	-
d	x	x	x	-	-	-	-
e	f-d b-c	d-d c-c	f-d e-c	x	-	-	-
f	x	x	x	x	x	-	-
g	x	x	x	x	x	x	-
h	x	x	x	g-g a-a	x	x	x
	a	b	c	d	e	f	g

- d. Draw the state diagram corresponding to the reduced state table.





- a. Derive the excitation equations, transition equations, and the output equations.

$$J_1 = Q_1 + Q_0$$

$$J_0 = Q_1$$

$$y = Q_1 + Q_0$$

$$K_1 = x$$

$$K_0 = x$$

Derivation:

$$Q_1(t+1) = J_1'Q_1' + K_1'Q_1'$$

$$Q_0(t+1) = J_0Q_0' + K_0'Q_0$$

$$Q_1(t+1) = (Q_0 + Q_0)Q_1' + x'Q_1 \quad Q_0(t+1) = Q_1Q_0' + x'Q_0$$

$$Q_1(t+1) = Q_0Q_1' + x'Q_1$$

Final equations:

$$Q_1(t+1) = Q_0Q_1' + x'Q_1,$$

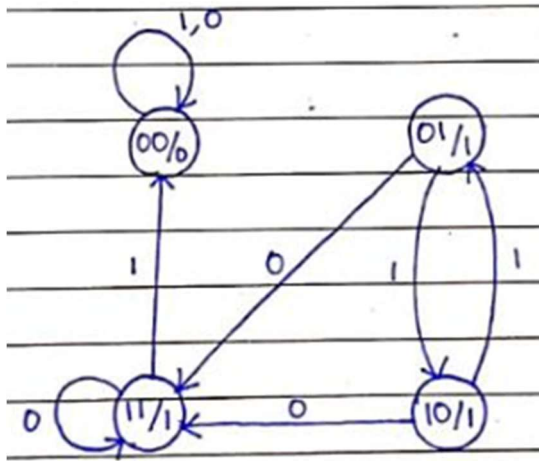
$$Q_0(t+1) = Q_1Q_0' + x'Q_0,$$

$$y = Q_1 + Q_0$$

- b. Create the state table that defines the current-states, input x, next-states, JK Flip Flop inputs (i.e., the excitation inputs), and the output y.

Current State		Inputs	Next State		Flip Flop Inputs				Output
Q_1	Q_0	x	Q_1^+	Q_0^+	J_1	K_1	J_0	K_0	y
0	0	0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0	1	0
0	1	0	1	1	1	0	0	0	1
0	1	1	1	0	1	1	0	1	1
1	0	0	1	1	1	0	1	0	1
1	0	1	0	1	1	1	1	1	1
1	1	0	1	1	1	0	1	0	1
1	1	1	0	0	1	1	1	1	1

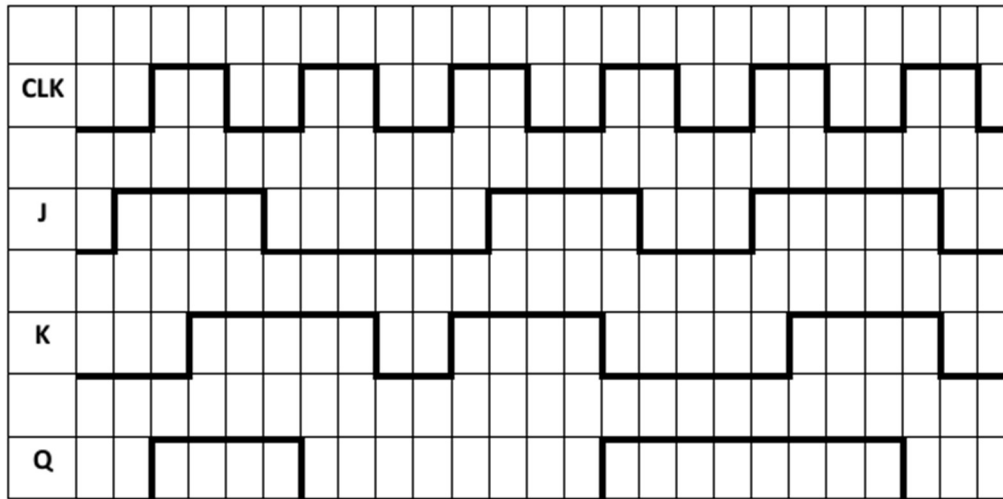
c. Create the state diagram and specify what type of state machine is this (Mealy or Moore).



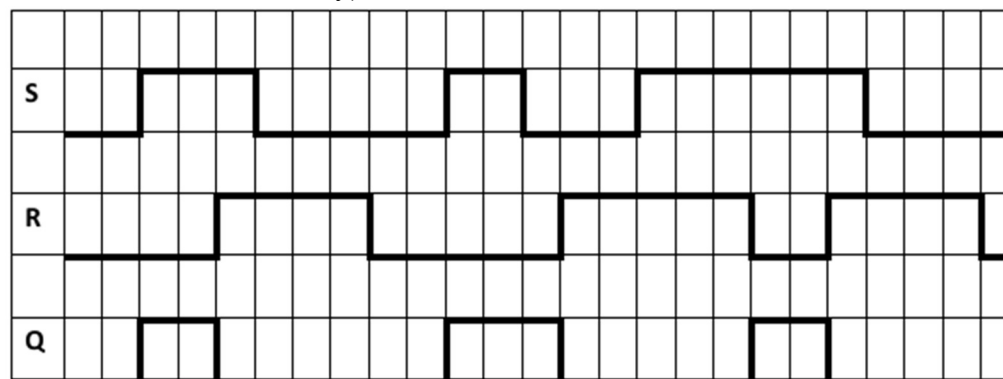
It is a Moore machine as the output only depends on the present states.

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a. Given the timing diagram of a positive edge triggered JK flip-flop, draw the output Q in the following timing diagram.

$$4x^2+10$$


b. The inputs for an SR-Latch are given in the following diagram. Complete the diagram by drawing the output Q. (Assume that output Q would be 0 initially)



- b. Draw the transition table for the circuit using T flip flops.

Current State	Next State
S ₀	S ₁
S ₁	S ₂
S ₂	S ₃
S ₃	S ₄
S ₄	S ₀

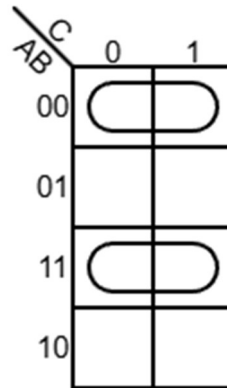
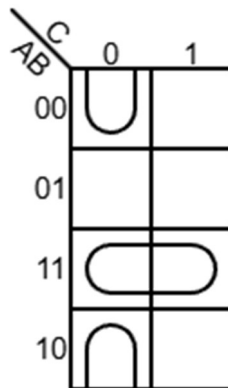
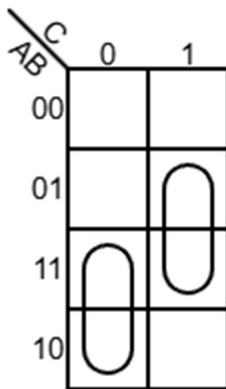
Three flip-flops (Q₂, Q₁, Q₀) are used to represent the states of a 5-state sequence.

For example, State S₀ = 000, where Q₂ = 0, Q₁ = 0, and Q₀ = 0, and the remaining states (S₁, S₂, S₃, S₄) follow accordingly with their respective binary representations.

Current State				Next State			Flip Flop Inputs		
	Q ₂	Q ₁	Q ₀	Q ₂ ⁺	Q ₁ ⁺	Q ₀ ⁺	D ₂	D ₁	D ₀
S ₀	0	0	0	0	1	1	0	1	1
	0	0	1	0	0	1	0	0	1
	0	1	0	0	1	0	0	1	0
S ₁	0	1	1	1	0	0	1	0	0
S ₂	1	0	0	1	1	0	1	1	0
S ₄	1	0	1	0	0	0	0	0	0
S ₃	1	1	0	1	0	1	1	0	1
	1	1	1	1	1	1	1	1	1

- c. Use K-maps to deduce the equations for each T input for the flip flops.

KMAPS

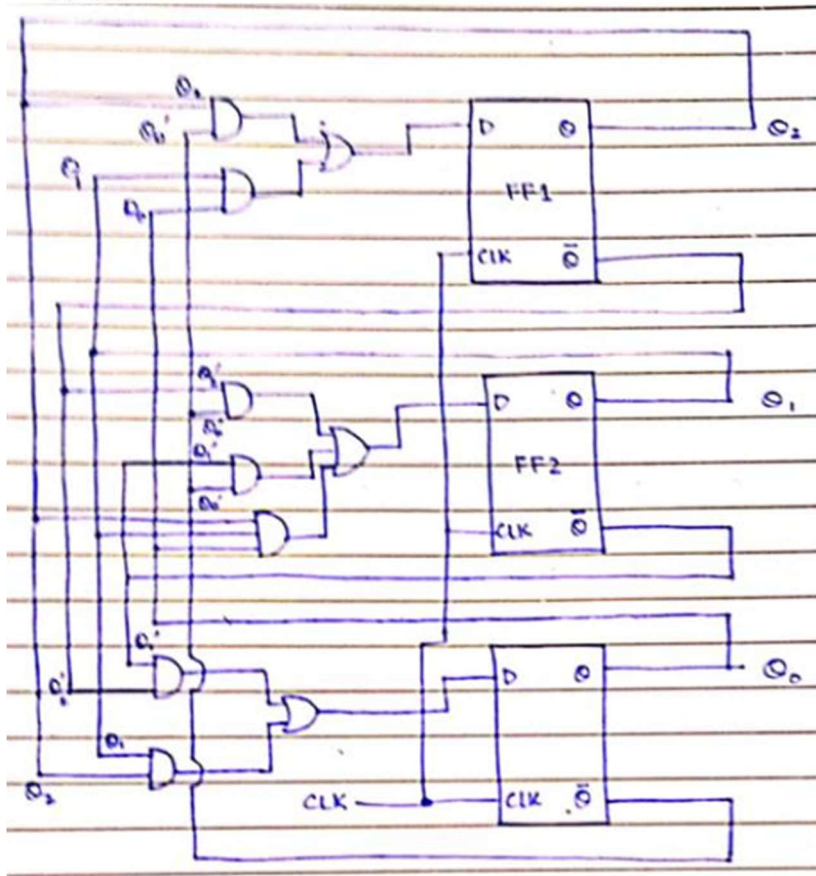


$$D_2 = Q_2Q_0' + Q_1Q_0$$

$$D_1 = Q_2Q_1 + Q_1'Q_0'$$

$$D_0 = Q_2'Q_1' + Q_2Q_1$$

d. Draw the circuit diagram for the sequential circuit.



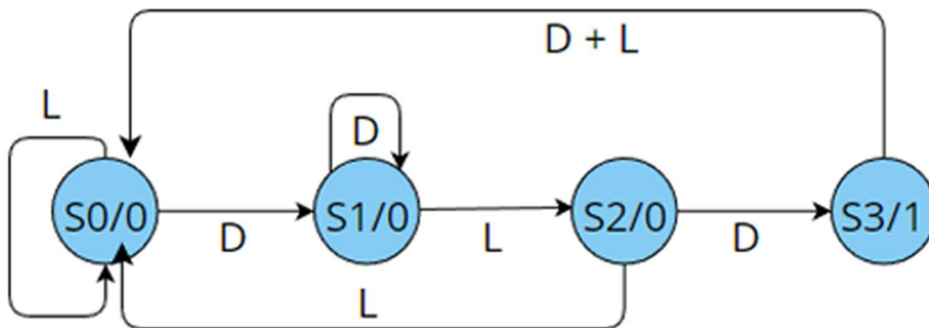
- 6 You decide to use a locker to store the equipment for your DLD project. Design a digital lock for your locker using JK flip-flop which opens if a person inputs the sequence "DLD". You can consider that you only have two inputs 'D' and 'L' and one output X (locked/ unlocked). Each element of the sequence must last for one or more cycles, and there may be zero or more cycles of both inputs low between the sequence elements. After the lock is open, either input going high causes the unlock to go low. Draw the circuit diagram for a digital lock implementation and show all the relevant working (state table, state diagram, K-maps, state equations).

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State Table:

Present State		Input		Next State		Output	FlipFlop A		FlipFlop B	
A	B	D	L	A	B	X	J _A	K _A	J _B	K _B
0	0	0	0	0	0	0	0	X	0	X
0	0	0	1	0	0	0	0	X	0	X
0	0	1	0	0	1	0	0	X	1	X
0	0	1	1	0	0	0	0	X	0	X
0	1	0	0	0	1	0	0	X	X	0
0	1	0	1	1	0	0	1	X	X	1
0	1	1	0	0	1	0	0	X	X	0
0	1	1	1	0	1	0	0	X	X	0
1	0	0	0	1	0	0	X	0	0	X
1	0	0	1	1	0	0	X	0	0	X
1	0	1	0	1	1	0	X	0	1	X
1	0	1	1	1	0	0	X	0	0	X
1	1	0	0	1	1	1	X	0	X	0
1	1	0	1	0	0	1	X	1	X	1
1	1	1	0	0	0	1	X	1	X	1
1	1	1	1	0	0	1	X	1	X	1

State Diagram:



Legend for States: S0 = 00, S1 = 01, S2 = 10, S3 = 11

KMAPS

Legend for States: S0 = 00, S1 = 01, S2 = 10, S3 = 11

X <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 0 01 0 0 0 0 11 1 1 1 1 10 0 0 0 0	J_a <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 0 01 0 1 0 0 11 - - - - 10 - - - -
J_b <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 0 0 0 1 01 - - - - 11 - - - - 10 - - - 1	K_a <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 - - - - 01 - - - - 11 0 1 1 1 10 0 0 0 0
K_b <i>D,L</i> 00 01 11 10 <i>A,B</i> 00 - - - - 01 0 1 0 0 11 0 1 1 1 10 - - - -	

State Equations for X, J_A, J_B, K_A and K_B:

$$X = AB$$

$$J_A = BD'L$$

$$J_B = D'L$$

$$K_A = BD + BL$$

$$K_B = D'L + AD$$

Circuit Diagram:

